Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/19/2012
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	Maturing Technology for Major Reductions in Energy, Mass, and Crew Time for Plant Lighting in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is now 7/31/2014 per NSSC information (Ed., 7/15/13) NOTE: new end date is 7/31/2013 per NSSC, and also is now showing start date of 6/9/2009 (Ed., 6/11/12) NOTE: new end date is 7/31/2012 per NSSC (Ed. 5/3/2011) NOTE: change in Period of Perfomance to 8/1/2009-7/31/2011 (from 6/9/2009-7/8/2011), per NSSC information (Ed., 3/11/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To develop and demonstrate proof of concept for two novel technologies that together will greatly advance the potential for growing food crops in space for crew life support at distant space destinations. The first technology is a “smart” LED-based sole-source lighting system for saving considerable energy for many reasons. The second technology is a crop-stand gas-exchange system consisting of a large cuvette, an open gas-exchange system, and computer controls for CO2, temperature, and atmosphere flow rate. Lighting parameters are controlled by a second computer. Using the smart lighting system in conjunction with the crop gas-exchange system will give real-time crop response of hydroponically grown crops to light (PPF, R/B ratio), temperature, and CO2 at all stages of crop production, which will define additional opportunities for significant energy savings during controlled-environment crop production.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project has broad impact for life on Earth as well as in space. On Earth, controlled environment agriculture (CEA) has limitations of profitability, economic viability, and affordability parallel to the issues that hold back the feasibility of using plants for crew life support in space. Interestingly, the single factor that limits the widespread use of CEA on Earth also is that of crop lighting! Supplemental and sole-source lighting using traditional high-intensity discharge (HID) lamps requires major electrical power consumption for waste-heat rejection as well as for lighting per se. The replacement of hot, inefficient HID lamps with solid-state light-emitting diode (LED) technology will benefit CEA and plant research for many reasons: LEDs are small, require much less mass and volume than HIDs, do not require massive ballasts, are extremely long-lived, are relatively cool at photon-emitting surfaces because waste heat is rejected elsewhere, emission wavelengths can be selected to match plant pigments, and can be operated at much lower power levels than HIDs because they can be placed in close proximity to plant tissues. As well, the field of LED technology is progressing at an ever-increasing rate, and electric-to-light conversion efficiencies are increasing at a rapid rate for LEDs. There is great interest in the use of LEDs for CEA on Earth as well as for crew life support in space. In addition, this project addresses the development of “smart” LED lighting systems that automatically sense plant position and size, and subsequently energize LEDs only directly above or in line with plant tissues. When used in conjunction with and supplemental to solar light in greenhouses on Earth, smart LED lighting systems also will monitor the instantaneous contribution of solar light as it tracks across greenhouses daily, and will automatically adjust LED switching and/or dimming/brightening controls in response to the total light environment. The crop gas-exchange system that is part of this project will be used in conjunction with smart LEDs to determine day-to-day crop tolerances, “preferences,” and limits for intensity of light, quality of light, and interaction of light with other important environmental parameters such as CO2 and temperature at all stages of crop development. This capability will greatly accelerate the optimization of environmental conditions needed to produce crops rapidly, efficiently, and affordably, which is just as important for CEA on Earth as it is for closed-loop life support in space.
Task Progress:	The first half of the reporting period involved troubleshooting and retrofitting of the smart LED HELIAC (high-efficiency lighting with adaptive control) hardware and software in consultation with ORBITEC engineers, including three trips to Madison, WI where ORBITEC is located, and one trip by ORBITEC engineers to Purdue. The second half of the reporting period involved experimentation with plants conducted in a walk-in growth growth room at Purdue University by the graduate student in Aerospace Engineering. Troubleshooting of the cuvette/gas-exchange system by the Electrical Engineering Technology student revealed the nature and location of a subtle design flaw in the crop cuvette that has been preventing the entire gas-exchange system from performing as per specifications. Specific retrofits included the following: addition of a dark-scan function to the software to automatically subtract false-positive images from array infrastructure obtained prior to planting plants below the array; removal of collimating lenses from red LEDs so that the array could be placed closer to the crop surface without losing spectral blending of red and blue radiation; repair of one quadrant’s energy-monitoring system on the LED array; replacement of several blue LEDs that had burned out. Experiments subsequently were conducted with the two refurbished HELIAC LED units to determine optimum red/blue ratios from LEDs during all stages of crop growth with leaf lettuce, which was the model crop used to determine proof of concept for smart lighting. Seedlings grown under lighting treatments with increasing blue light developed progressively shorter hypocotyls (lower stems), and the cotyledons (seed leaves) and true leaves were less expanded. It also was observed that seedlings becoming excessively elongated early on (due to low total PPF (photosynthetic photon flux) and/or low B/R ratio did not grow well subsequently: small leaves, low fresh and dry weights, and frequently stem breakage leading to seedling death. Subsequent experiments defined a lighting protocol for efficient seedling establishment, to find the best lighting environment in terms of blue/red ratio to get a healthy seedling (hypocotyl not too long, large leaf area and adequate fresh/dry weight). Six different fairly low light-intensities were tested over a 9-day slow-growth (lag phase) period during which the seedling is getting geared up for rapid growth. For the same low intensity, one delivered 5% blue light; another 10% of blue light; one 5% of blue light for the first two days, and 10% blue the remaining 7 days; one with 10% blue for the first 2 days, and 5% blue for the remaining 7 days; one with 100% red light for the first 2 days, and 5% blue the remaining 7 days; one with 100% red the first 2 days, and 10% blue light the remaining 7 days. Giving 5% blue light over the entire lag phase was the optimal red/blue ratio. Continuous red light during the first 2 days after emergence didn’t result in optimum growth because hypocotyl lengths were too long. Seedlings under higher blue light developed shorter hypocotyls and smaller leaf area. The first 2 days after planting were critical for seedling establishment. Some seedlings suffered from epinasty, a downward bending and curling of leaves under too much light. Thus, light was increased incrementally by adding it in daily, small steps. This stepwise increase also was better in terms of energy savings. Transition of the timing from lag (slow) to exponential (fast) seedling growth was determined by periodic harvest of plants from the hydroponics system over a 26-day growth period. Growth under LEDs was identical for plants grown under 10% blue light vs. 5% blue. From day15, the onset of exponential growth, to day 25, energy expenditure per unit of dry edible lettuce biomass was 0.39 kWh/g DW. An experiment was conducted comparing the energy efficiency of an LED array in which all LEDS were energized during each light period of crop growth vs. an array where LEDs were selectively energized only directly above each seedling, and as seedlings grew in size and diameter, more and more LEDs were energized until seedlings had grown together and all seedlings were energized. The energy required to grow shoot biomass for the all-on control was 1.39 kWh/g, while that of the selectively switched-on array was 1.14 kWh/g DW shoot biomass. The Engineering Technology student working with the cuvette designer tested the gas-exchange system (infrared gas

	analyzers, mass-flow controllers, control computer) using a 6-gallon sealed bucket fitted with an inlet and outlet port as a surrogate test unit for the larger crop cuvette with suspected design flaws. The objective of the test was to demonstrate that a cuvette containing no plants should have the same or similar CO2 concentration exiting the outlet port of the empty chamber as entered the inlet port. A second gas stream bypasses the cuvette entirely and serves as a reference for the gas stream passing through the cuvette. The analytical system performed within specifications over a range of elevated CO2 concentrations, plus responded with a CO2 differential when an NAOH CO2 trap was interduced into the test chamber, indicating that it was operating properly and that the source of error was indeed the larger cuvette. When test air is run through the larger crop cuvette with temperature-control running but no plants present, CO2 differentials as high as -150 to -200 PPM are detected, indicating flaws related to unprocessed inlet air plus cooling of outlet air to below the equilibrium temperature. Prior to retrofitting the larger crop cuvette, a simplified cuvette system is being designed and fabricated that has a much smaller head-space volume than the original cuvette. This snap-on/clamp-on cuvette will fit over the hydroponic tray and around one of the HELIAC LED arrays. The HELIAC/cuvette/hydroponic unit will be deployed in a walk-in growth chamber, with the hydroponic line and pulley system mounted above the table from unistrut infrastructure, and the clamshell cuvette supported, in part, from below by the hydroponics lid it is resting on and attached to, and, in part, from above, by the HELIAC unit to which it is sealed. A manifold at the inlet end of the cuvette will distribute the inlet gas stream uniformly above and across rows of growing lettuce seedlings and below the HELIAC LED array, which will be deployed about 6 cm above the lettuce canopy and will be readjusted daily to maintain that separation distance as plants grow in height as well as in diameter. Air moving across to the far end of the cuvette will be funneled to an outlet port and airline that will be directed to the IRGAs located outside the growth chamber. The Peltier unit that was obtained for the crop cuvette will be interfaced with the cuvette within the growth room to maintain an equilibrium cuvette temperature with LEDs energized. The setpoint temperature will be controlled by the gas-exchange control computer. Waste heat from the HELIAC LED array will be vented from the top of the array into the controlled-environment room, so the operational cuvette heat load on the Peltier unit should be nominal. Temperature in the controlled-environment room also will help to maintain cuvette temperature with the LEDs energized. Design and initiation of fabrication of the clamp-on cuvette were initiated during the reporting period. The Engineering Technology student also updated a set of instructions originally created by a predecessor regarding operation of the cuvette/gas-exchange system for the benefit of future users.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Poulet L, Massa G, Bourget M, Morrow R, Mitchell C. "Smart lighting for major reductions in energy for plant lighting in space." Presented at the 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 87. , Nov-2011